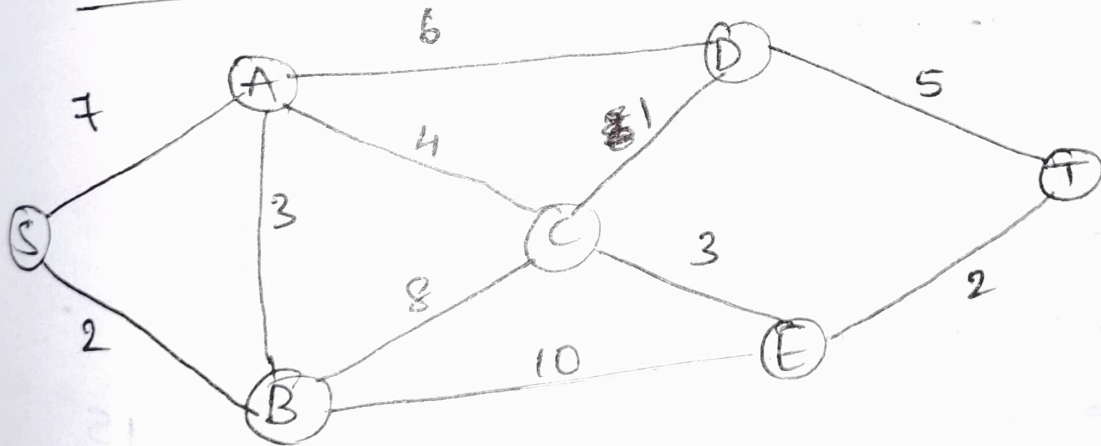


Exercise-2 :- shortest paths from S



Node	cost	previous
S	0	-
A	∞ 7/5	S/B
B	∞ 2	S
C	∞ 10/9	B/A
D	∞ 11/10	A/C
E	∞ 12	B
T	∞ 15/14	D/E

if $d(u) + c(u, v) < d(v)$
then update the table.

(A, 7) (A, 5) 2	(B, 2) 1	(C, 10) (C, 9) 3	(E, 12) 5	(D, 11) (D, 10) 4	(T, 15) (T, 14) 6
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priority queue

at S:- $d(S) = 0$

$S \rightarrow A$: $0 + 7 < \infty = 7 < \infty \checkmark$ then update $d(A) = 7$

$S \rightarrow B$: $0 + 2 < \infty = 2 < \infty \checkmark$ then update $d(B) = 2$

at B:- $d(B) = 2$

$B \rightarrow A$: $2 + 3 < 7 = 5 < 7 \checkmark$ then update $d(A) = 5$

$B \rightarrow S$: $2 + 2 < 0 = 4 < 0 \otimes$ do not update.

$B \rightarrow C$: $2 + 8 < \infty = 10 < \infty \checkmark$ update $d(C) = 10$

$B \rightarrow E$: $2 + 10 < \infty = 12 < \infty \checkmark$ update $d(E) = 12$

at A:- $d(A) = 5$

$A \rightarrow B$: $5 + 3 < 2 = 8 < 2 \otimes$ do not update

$A \rightarrow S$: $5 + 7 < 0 = 12 < 0 \otimes$ do not update

$A \rightarrow C$: $5 + 4 < 10 = 9 < 10 \checkmark$ then update $d(C) = 9$

$A \rightarrow D$: $5 + 6 < \infty = 11 < \infty \checkmark$ update $d(D) = 11$

at C:- $d(C) = 9$

$C \rightarrow A$: $9 + 4 < 5 = 13 < 5 \otimes$

$C \rightarrow B$: $9 + 8 < 2 = 17 < 2 \otimes$

$C \rightarrow D$: $9 + 1 < 11 = 10 < 11 \checkmark$ update $d(D) = 10$

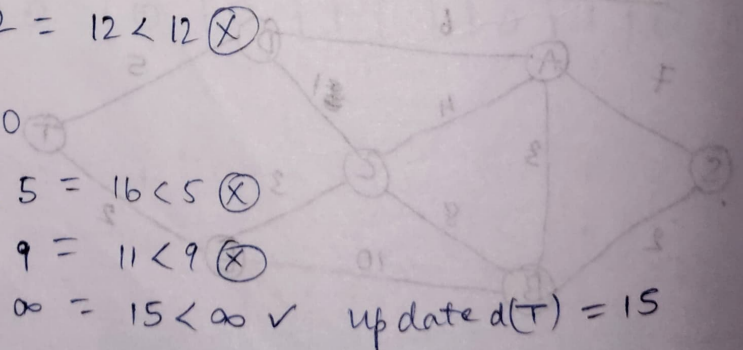
$C \rightarrow E$: $9 + 3 < 12 = 12 < 12 \otimes$

at D:- $d(D) = 10$

$D \rightarrow A$: $10 + 6 < 5 = 16 < 5 \otimes$

$D \rightarrow C$: $10 + 1 < 9 = 11 < 9 \otimes$

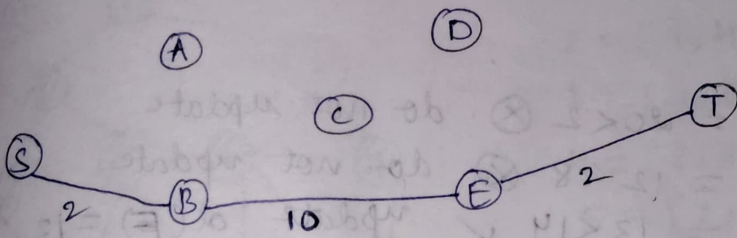
$D \rightarrow T$: $10 + 5 < \infty = 15 < \infty \checkmark$ update $d(T) = 15$



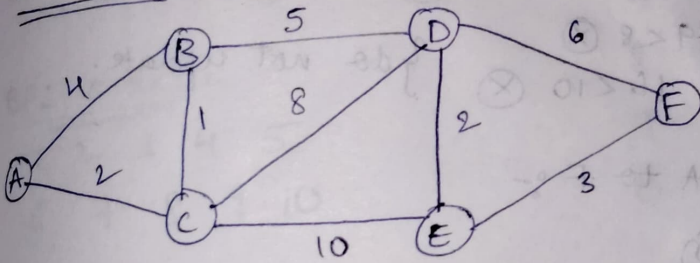
$d(E) = 2$
 $E \rightarrow C: 12 + 3 < 9 = 15 < 9 \times$
 $E \rightarrow B: 12 + 10 < 2 = 22 < 2 \times$
 $E \rightarrow T: 12 + 2 < 15 = 14 < 15 \checkmark$ update $d(T) = 14$
 at $T: d(T) = 14$

$T \rightarrow D: 14 + 5 < 10 \times$ do not update
 $T \rightarrow E: 14 + 2 < 12 \times$ do not update.

now from S to T. :-



Exercise-1 :- shortest paths from A



Node	cost	previous
A	0	-
B	4 3	A, C
C	2	A
D	10 8	C, B
E	12 10	C, D
F	14 13	D, E

priority Queue

(B, 4)	(C, 2)	(E, 12)	(D, 10)	(F, 14)	
(B, 3)	D	(E, 10)	(D, 8)	(F, 13)	
2		7	3	5	

at A :- $d(A) = 0$
 $A \rightarrow B: 0 + 4 < \infty = 4 < \infty \checkmark$ update $d(B) = 4$
 $A \rightarrow C: 0 + 2 < \infty = 2 < \infty \checkmark$ update $d(C) = 2$

at C :- $d(C) = 2$
 $C \rightarrow A: 2 + 2 < 0 = 4 < 0 \times$ updation not done
 $C \rightarrow E: 2 + 10 < \infty = 12 < \infty \checkmark$ update $d(E) = 12$
 $C \rightarrow D: 2 + 8 < \infty = 10 < \infty \checkmark$ update $d(D) = 10$

$C \rightarrow B: 2+1 < 4 = 3 < 4 \checkmark$ update $d(B) = 3$

at B $\therefore d(B) = 3$

$B \rightarrow A: 3+4 < 0 = 7 < 0 \times$ do not update

$B \rightarrow C: 3+1 < 2 = 4 < 2 \times$ do not update

$B \rightarrow D: 3+5 < 10 = 8 < 10 \checkmark$ update $d(D) = 8$

at D $\therefore d(D) = 8$

$D \rightarrow B: 8+5 < 3 = 13 < 3 \times$

do not update

$D \rightarrow C: 8+8 < 2 = 16 < 2 \times$

do not update

$D \rightarrow E: 8+2 < 12 = 10 < 12 \checkmark$

update $d(E) = 10$

$D \rightarrow F: 8+6 < \infty = 14 < \infty \checkmark$

update $d(F) = 14$

at E $\therefore d(E) = 10$

$E \rightarrow C: 10+10 < 2 = 20 < 2 \times$ do not update

$E \rightarrow D: 10+2 < 8 = 12 < 8 \times$ do not update

$E \rightarrow F: 10+3 < 14 = 13 < 14 \checkmark$ update $d(F) = 13$

at F $\therefore d(F) = 13$

$F \rightarrow D: 13+6 < 8 = 19 < 8 \times$

$F \rightarrow E: 13+3 < 10 = 16 < 10 \times$ } do not update.

Now from A to F:-

